Exam Formula Sheet

1 Epi 202: Probability

$$
\begin{aligned}
\operatorname{Var}\left(\tilde{a} \cdot \tilde{X}\right) &= \operatorname{Var}\left(\sum_{i=1}^{n} a_i X_i\right) \\
&= \tilde{a}^{\top} \operatorname{Var}\left(\tilde{X}\right) \tilde{a} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j \operatorname{Cov}\left(X_i, X_j\right)
\end{aligned}
$$

$$\mathrm{E}[Y] = \mathrm{E}[\mathrm{E}[Y \mid X]]$$

$$\mathrm{E}[Y \mid Z] = \mathrm{E}[\mathrm{E}[Y \mid X, Z] \mid Z]$$

$$\operatorname{Var}(Y) = \mathrm{E}[\operatorname{Var}(Y \mid X)] + \operatorname{Var}(\mathrm{E}[Y \mid X])$$

$$\operatorname{Cov}(Y, Z) = \mathrm{E}[\operatorname{Cov}(Y, Z \mid X)] + \operatorname{Cov}(\mathrm{E}[Y \mid X], \mathrm{E}[Z \mid X])$$

2 Epi 203: Statistical inference

$$\mathcal{L}(\theta) \stackrel{\text{def}}{=} \mathrm{p}(\tilde{X} = \tilde{x} | \Theta = \theta)$$

$$\ell \stackrel{\text{def}}{=} \log\{\mathcal{L}(\tilde{x}|\theta)\}$$

$$\ell' \stackrel{\text{def}}{=} \frac{\partial}{\partial \theta} \ell(\tilde{x}|\theta)$$

$$\ell'' \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{\theta}} \frac{\partial}{\partial \tilde{\theta}^{\top}} \ell(\tilde{x}|\tilde{\theta})$$

$$\ell''_{ij} = \frac{\partial}{\partial \theta_i} \frac{\partial}{\partial \theta_j} \ell(\tilde{X} = \tilde{x}|\tilde{\theta})$$

$$
\begin{aligned}
I &\stackrel{\text{def}}{=} -\ell''(\tilde{x}|\tilde{\theta}) \\
\mathcal{I} &\stackrel{\text{def}}{=} \mathrm{E}[I(\tilde{x}|\theta)]
\end{aligned}
$$

$$\hat{\theta}_{ML} \dot{\sim} \mathrm{N}\left(\theta, \left[\mathcal{I}(\tilde{\theta})\right]^{-1}\right)$$

For one parameter θ_k:

$$\mathrm{CI}_{1-\alpha}(\theta_k) = \left[\hat{\theta}_k \pm z_{1-\frac{\alpha}{2}} \widehat{\mathrm{SE}}\left(\hat{\theta}_k\right)\right]$$

$$Z_k \stackrel{\text{def}}{=} \frac{\hat{\theta}_k - \theta_{k,0}}{\widehat{\mathrm{SE}}\left(\hat{\theta}_k\right)} \dot{\sim} \mathrm{N}(0,1) \qquad \text{under } H_0: \theta_k = \theta_{k,0} \qquad z_k = \text{observed } Z_k$$

$$\begin{aligned} p\text{-value} &= 2\Pr\left(|Z| \geq |z_k|\right), \quad Z \sim \mathrm{N}(0,1) \\ &= 2(1 - \Phi(|z_k|)) \end{aligned}$$

3 Sta 108: Linear regression

$$t_k \stackrel{\text{def}}{=} \frac{\hat{\beta}_k - \beta_{k,0}}{\widehat{\mathrm{SE}}\left(\hat{\beta}_k\right)} \dot{\sim} t_{n-p} \qquad \text{under } H_0: \beta_k = \beta_{k,0} \qquad t_k^{\text{obs}} = \text{observed } t_k$$

$$\mathrm{CI}_{1-\alpha}(\beta_k) = \left[\hat{\beta}_k \pm t_{n-p}\left(1 - \frac{\alpha}{2}\right)\widehat{\mathrm{SE}}\left(\hat{\beta}_k\right)\right]$$

Let $T_{n-p} \sim t_{n-p}$.

$$\begin{aligned} p\text{-value} &= 2\Pr\left(\left|T_{n-p}\right| \geq \left|t_k^{\text{obs}}\right|\right) \\ &= 2\left(1 - F_{t_{n-p}}\left(\left|t_k^{\text{obs}}\right|\right)\right) \end{aligned}$$

$$\mathrm{CI}_{1-\alpha}(\mu(\tilde{x}^*)) = \left[\hat{\mu}(\tilde{x}^*) \pm t_{n-p}\left(1 - \frac{\alpha}{2}\right)\widehat{\mathrm{SE}}(\hat{\mu}(\tilde{x}^*))\right]$$

Y^* denotes a new observation (not in the training data), with corresponding covariate pattern $\tilde{x}^*$. Let $\hat{Y}^* \stackrel{\text{def}}{=} \hat{\mu}(\tilde{x}^*)$.

$$\mathrm{PI}_{1-\alpha}(Y^*|\tilde{x}^*) = \left[\hat{Y}^* \pm t_{n-p}\left(1 - \frac{\alpha}{2}\right)\widehat{\mathrm{SE}}\left(Y^* - \hat{Y}^*\right)\right]$$

$$\widehat{\mathrm{SE}}\left(Y^* - \hat{Y}^*\right) = \hat{\sigma}\sqrt{1 + (\tilde{x}^*)^\top (\mathbf{X}'\mathbf{X})^{-1}\tilde{x}^*}$$

$$\mathrm{Var}\left(Y^* - \hat{Y}^*\right) = \sigma^2\left(1 + (\tilde{x}^*)^\top (\mathbf{X}'\mathbf{X})^{-1}\tilde{x}^*\right)$$

Let $\hat{\Sigma} \stackrel{\text{def}}{=} \widehat{\mathrm{Var}}\left(\hat{\beta}\right) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$.

$$\widehat{\mathrm{Var}}(\hat{\mu}(\tilde{x})) = \tilde{x}^\top \hat{\Sigma} \tilde{x}$$

Let $\Delta\mu(\tilde{x}, \tilde{x}^*) = \mu(\tilde{x}) - \mu(\tilde{x}^*)$, and let $\Delta\tilde{x} = \tilde{x} - \tilde{x}^*$; then:

$$\widehat{\mathrm{Var}}\left(\widehat{\Delta\mu}(\tilde{x}, \tilde{x}^*)\right) = \Delta\tilde{x}^\top \hat{\Sigma} \Delta\tilde{x}$$

4 Epi 204: Generalized linear models

Generalized linear models have three components:

1. The **outcome distribution** family: $\mathrm{p}(Y|\mu(\tilde{x}))$
2. The **link function**: $g(\mu(\tilde{x})) = \eta(\tilde{x})$
3. The **linear component**: $\eta(\tilde{x}) = \tilde{x} \cdot \beta$

$$\left[\pi \overset{\text{def}}{=} \Pr(Y=1|\tilde{X}=\tilde{x})\right] \overbrace{\underbrace{\underset{\frac{\omega}{1+\omega}}{\overset{\frac{\pi}{1-\pi}}{\rightleftarrows}} \left[\omega \overset{\text{def}}{=} \text{odds}(Y=1|\tilde{X}=\tilde{x})\right] \underset{\exp\{\eta\}}{\overset{\log\{\omega\}}{\rightleftarrows}}}_{\text{expit}(\eta)}}^{\text{logit}(\pi)} \left[\eta(\tilde{x}) \overset{\text{def}}{=} \text{log-odds}(Y=1|\tilde{X}=\tilde{x})\right]$$

Figure 1: Diagram of logistic regression link and inverse link functions

$$\theta(\tilde{x}, \tilde{x}^*) = \exp\left\{(\Delta\tilde{x}) \cdot \tilde{\beta}\right\}$$

4.1 Estimates of odds ratios from 2x2 contingency tables

$$\hat{\theta} = \frac{ad}{bc}$$

4.2 Survival analysis

4.2.1 Probability distribution functions

Table 1: Probability distribution functions

Name	Symbols	Definition
Probability density function (PDF)	$\mathrm{f}(t), \mathrm{p}(t)$	$\mathrm{p}(T = t)$
Cumulative distribution function (CDF)	$\mathrm{F}(t), \mathrm{P}(t)$	$\mathrm{P}(T \leq t)$
Survival function	$\mathrm{S}(t), \bar{\mathrm{F}}(t)$	$\mathrm{P}(T > t)$
Hazard function	$\lambda(t), \mathrm{h}(t)$	$\mathrm{p}(T = t \mid T \geq t)$
Cumulative hazard function	$\Lambda(t), \mathrm{H}(t)$	$\int_{u=-\infty}^{t} \lambda(u)du$
Log-hazard function	$\eta(t)$	$\log\{\lambda(t)\}$

4.2.2 Diagram of survival distribution function relationships

$$\mathrm{f}(t) \xleftarrow[\mathrm{S}(t)\lambda(t)]{-S'(t)} \mathrm{S}(t) \xleftarrow{\exp\{-\Lambda(t)\}} \Lambda(t) \xleftarrow{\int_{u=0}^{t} \lambda(u)du} \lambda(t) \xleftarrow{\exp\{\eta(t)\}} \eta(t)$$

$$\mathrm{f}(t) \xrightarrow[\int_{u=t}^{\infty} \mathrm{f}(u)du]{\mathrm{f}(t)/\lambda(t)} \mathrm{S}(t) \xrightarrow[-\log \mathrm{S}(t)]{} \Lambda(t) \xrightarrow[\Lambda'(t)]{} \lambda(t) \xrightarrow[\log\{\lambda(t)\}]{} \eta(t)$$

4.2.3 Survival likelihood contributions, assuming non-informative censoring

$$\begin{aligned}\mathrm{p}(Y = y, D = d) &= [\mathrm{f}_T(y)]^d [\mathrm{S}_T(y)]^{1-d} \\ &= [\lambda_T(y)]^d [\mathrm{S}_T(y)]\end{aligned}$$

4.2.4 Nonparametric time-to-event distribution estimators

$$\hat{\lambda}_i = \frac{d_i}{n_i}$$

$$\hat{\mathrm{S}}_{KM}(t) \stackrel{\text{def}}{=} \prod_{\{i:\ t_i < t\}} \left[1 - \hat{\lambda}_i\right]$$

$$\hat{\Lambda}_{NA}(t) \stackrel{\text{def}}{=} \sum_{\{i:\ t_i < t\}} \hat{\lambda}_i$$

4.2.5 Proportional hazards model structure

Joint likelihood of data set: $\mathcal{L} \stackrel{\text{def}}{=} \text{p}(\tilde{Y} = \tilde{y}, \tilde{D} = \tilde{d} | \mathbf{X} = \mathbf{x})$

Marginal likelihood contribution of obs. $\boldsymbol{i}$: $\mathcal{L}_i \stackrel{\text{def}}{=} \text{p}(Y_i = y_i, D_i = d_i | \tilde{X}_i = \tilde{x}_i)$

Independent Observations Assumption: $\mathcal{L} = \prod_{i=1}^{n} \mathcal{L}_i$

Non-Informative Censoring Assumption: $T_i \perp\!\!\!\perp C_i | \tilde{X}_i$

$$\mathcal{L}_i \propto [\text{f}_T(y_i|\tilde{x}_i)]^{d_i} [\text{S}_T(y_i|\tilde{x}_i)]^{1-d_i} = \text{S}_T(y_i|\tilde{x}_i) \cdot [\lambda_T(y_i|\tilde{x}_i)]^{d_i}$$

Survival function: $\text{S}(t|\tilde{x}) \stackrel{\text{def}}{=} \text{P}(T > t | \tilde{X} = \tilde{x}) = \int_{u=t}^{\infty} \text{f}(u|\tilde{x}) du = \exp\{-\Lambda(t|\tilde{x})\}$

Probability density function: $\text{f}(t|\tilde{x}) \stackrel{\text{def}}{=} \text{p}(T = t | \tilde{X} = \tilde{x}) = -\text{S}'(t|\tilde{x}) = \lambda(t|\tilde{x}) \text{S}(t|\tilde{x})$

Cumulative hazard function: $\Lambda(t|\tilde{x}) \stackrel{\text{def}}{=} \int_{u=0}^{t} \lambda(u|\tilde{x}) du = -\log\{\text{S}(t|\tilde{x})\}$

Hazard function: $\lambda(t|\tilde{x}) \stackrel{\text{def}}{=} \text{p}(T = t | T \geq t, \tilde{X} = \tilde{x}) = \Lambda'(t|\tilde{x}) = \frac{\text{f}(t|\tilde{x})}{\text{S}(t|\tilde{x})}$

Hazard ratio: $\theta(t|\tilde{x} : \tilde{x}^*) \stackrel{\text{def}}{=} \frac{\lambda(t|\tilde{x})}{\lambda(t|\tilde{x}^*)}$

Log-Hazard function: $\eta(t|\tilde{x}) \stackrel{\text{def}}{=} \log\{\lambda(t|\tilde{x})\} = \eta_0(t) + \Delta\eta(t|\tilde{x})$

Proportional Hazards Assumption:

$$\begin{aligned}
\lambda(t|\tilde{x}) &= \lambda_0(t) \cdot \theta(\tilde{x}) \\
\Lambda(t|\tilde{x}) &= \Lambda_0(t) \cdot \theta(\tilde{x}) \\
\eta(t|\tilde{x}) &= \eta_0(t) + \Delta\eta(\tilde{x})
\end{aligned}$$

Logarithmic Link Function Assumption:

- **Link function**:

$$\log\{\lambda(t|\tilde{x})\} = \eta(t|\tilde{x})$$

$$\log\{\theta(\tilde{x})\} = \Delta\eta(\tilde{x})$$

- **Inverse link function**:

$$\lambda(t|\tilde{x}) = \exp\{\eta(t|\tilde{x})\}$$

$$\theta(\tilde{x}) = \exp\{\Delta\eta(\tilde{x})\}$$

Linear Predictor Component:

$$\eta(t|\tilde{x}) = \eta_0(t) + \Delta\eta(t|\tilde{x})$$

$$\Delta\eta(t|\tilde{x}) = \tilde{x} \cdot \tilde{\beta}$$

Linear Predictor Component Functional Form Assumption:

$$\Delta\eta(t|\tilde{x}) = \tilde{x} \cdot \tilde{\beta} \stackrel{\text{def}}{=} \beta_1 x_1 + \cdots + \beta_p x_p$$

4.2.6 Proportional hazards model partial likelihood formula:

$$\mathcal{L}_i^* = \frac{\theta(\tilde{x}_i)}{\sum_{k \in R(t_i)} \theta(\tilde{x}_k)}$$
$$\mathcal{L}^* = \prod_{\{i:\ d_i = 1\}} \mathcal{L}_i^*$$

4.2.7 Proportional hazards model baseline cumulative hazard estimator:

$$\hat{\Lambda}_0(t) = \sum_{t_i < t} \frac{d_i}{\sum_{k \in R(t_i)} \theta(x_k)}$$